

1 §1.1 Population Models



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 5

§1.1 Population Models

Thursday 17 September 2026

Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, after class.

Population Models

Let P denote **population density** e.g., number of bacteria per unit area on a surface.

Suppose the population density grows in proportion to the current population density, i.e.,

$$\frac{dP}{dt} = rP,$$

where r is a positive constant (the **growth rate per capita**).

If there are any individuals at all, then $P > 0$.

$$\begin{aligned} P > 0 &\implies rP > 0 \implies \frac{dP}{dt} > 0 \implies P \text{ always increasing} \\ &\implies \frac{1}{P} \text{ always well-defined} \implies \frac{1}{P} \frac{dP}{dt} = r \end{aligned}$$

Population Models

Questions to think about:

- How can we solve the IVP: $\frac{1}{P} \frac{dP}{dt} = r$ subject to $P(0) = P_0$?
- Is this a good model for population growth? If not, why not?
- How can we improve the model?
- What is the phase line for this model? for your better model?
- What can we conclude about long term population dynamics from the phase line?

Population Models

Solving the *Malthusian model* $\frac{dP}{dt} = rP$:

$$\frac{1}{P} \frac{dP}{dt} = r \quad \Rightarrow \quad \int_0^t \frac{1}{P} \frac{dP}{dt} dt = \int_0^t r dt$$

$$\Rightarrow \quad \int_{P(0)}^{P(t)} \frac{1}{P} dP = \int_0^t r dt$$

$$\Rightarrow \quad \log |P| \Big|_{P(0)}^{P(t)} = rt \Big|_0^t$$

$$\Rightarrow \quad \log |P(t)| - \log |P(0)| = r(t - 0)$$

$$\Rightarrow \quad \log |P(t)/P(0)| = rt$$

$$\Rightarrow \quad |P(t)/P(0)| = e^{rt}$$

$$\Rightarrow \quad |P(t)| = |P(0)| e^{rt} = P(0)e^{rt} \quad \because P(0) > 0$$

$$\Rightarrow \quad P(t) = P(0)e^{rt}$$

Population Models

Malthusian model

$$\text{IVP: } \frac{dP}{dt} = rP, \quad P(0) = P_0 \quad \implies \quad P(t) = P_0 e^{rt}$$

Is this a good model of population growth?

- No population can grow exponentially forever.
- Rate of growth *per capita* is **not** constant: depends on density
- At higher density, growth is slower: crowding
- If density is sufficiently high, population will decline (growth rate will be negative)

Population Models

- Constant *per capita* growth rate is absurd in the long term
- Simplest model of density-dependent *per capita* growth rate:

$$r\left(1 - \frac{P}{K}\right)$$

where $r > 0$ and $K > 0$ are constants.

- r now means the ***intrinsic growth rate*** (*per capita*) or ***unconstrained growth rate*** (*per capita*).
 - If the population density P is very small ($P \ll K$) then *per capita* growth will be unconstrained, i.e., $dP/dt \approx rP$.
- K represents the ***carrying capacity*** of the environment
 - $0 < P < K \implies$ population grows
 - $P > K \implies$ population declines
 - $P = K \implies$ equilibrium (***stable***)

Logistic Model

The logistic model is sometimes written in expanded form:

$$\frac{dP}{dt} = aP - bP^2, \quad a > 0, \quad b > 0.$$

To identify the intrinsic growth rate r and carrying capacity K , factor the right side:

$$\begin{aligned} aP - bP^2 &= aP \left(1 - \frac{b}{a}P \right) \\ &= aP \left(1 - \frac{P}{a/b} \right). \end{aligned}$$

Therefore

$$r = a, \quad K = \frac{a}{b}.$$

The nonzero equilibrium is consequently $P = a/b = K$.

Logistic Model

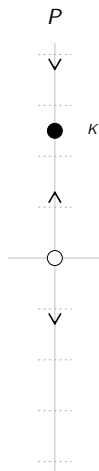
The *logistic model* of population growth

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

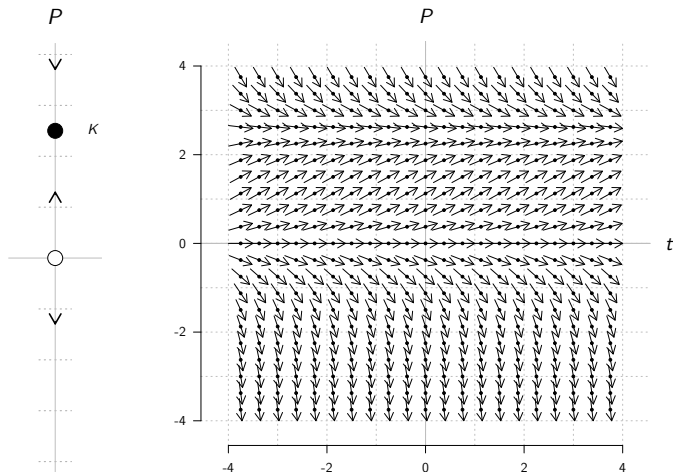
What are the equilibria of this autonomous ODE?

$$\frac{dP}{dt} = 0 \implies P = 0 \quad \text{or} \quad P = K$$

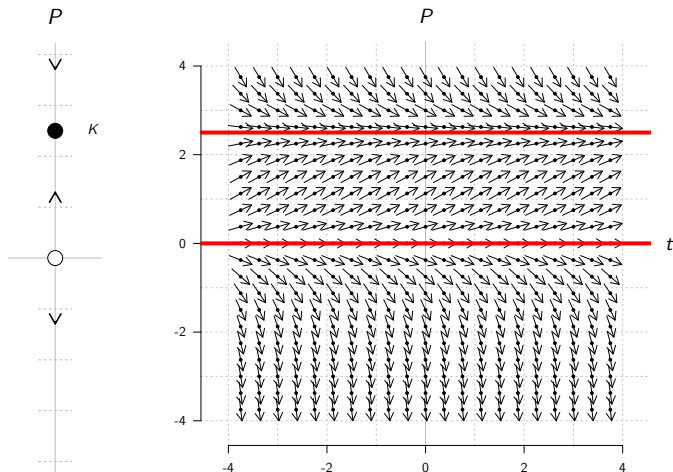
How can we display the solutions qualitatively on the phase line?



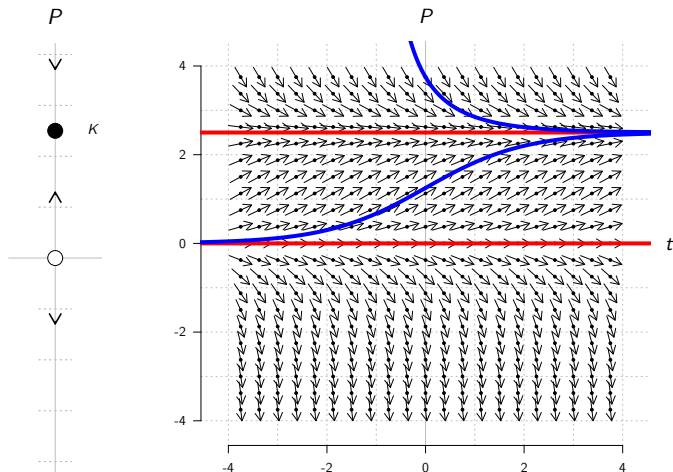
Logistic Model: direction field



Logistic Model: equilibria



Logistic Model: equilibria and other solution curves



Logistic Model: exact solution?

Consider the initial value problem:

$$\text{Solve } \frac{dP}{dt} = r P \left(1 - \frac{P}{K}\right) \quad \text{subject to} \quad P(0) = P_0$$

Is there always a unique solution to this initial value problem?

For any $P_0 \in \mathbb{R}$? For all time $t \geq 0$?

Can we find an exact formula for $P(t)$?

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 2C03**
- Click on **Take Class Poll**
- Fill in poll **Logistic-model IVP**
- .

Logistic Model: exact solution?

Consider the initial value problem:

$$\text{Solve } \frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right) \quad \text{subject to } P(0) = P_0$$

There is a unique solution to this IVP...

- for any $P_0 \geq 0$
 - YES.
- for any $P_0 \in \mathbb{R}$
 - YES, *but the solution is biologically meaningless if $P_0 < 0$.*
- for $t \in (-h, h)$ for some $h > 0$
 - YES, *follows from existence-uniqueness theorem*
- for all $t \geq 0$
 - YES, *provided $P_0 \geq 0$, but does not follow from theorem we discussed*
- for $t \in (-\infty, \infty)$
 - YES *provided $0 \leq P_0 \leq K$;*
 - NO, *otherwise* (not obvious!) We'll see why in a few minutes...

Logistic Model: exact solution?

Consider the initial value problem:

$$\text{Solve } \frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right) \quad \text{subject to } P(0) = P_0$$

Can we find an exact solution? *i.e.*, a formula for $P(t)$?

YES.

How?...

We need all the P 's on one side of the equation.

But we cannot divide by zero.

- **Case 1:** $P_0 = 0 \implies$ unique solution is $P(t) \equiv 0$
- **Case 2:** $P_0 = K \implies$ unique solution is $P(t) \equiv K$
- **Case 3:** $P_0 > 0$ and $P_0 \neq K \implies \frac{1}{P(1 - \frac{P}{K})} \frac{dP}{dt} = r$

Logistic Model: exact solution

$$P_0 > 0 \text{ and } P_0 \neq K \implies r = \frac{1}{P(1 - \frac{P}{K})} \frac{dP}{dt}$$

$$\implies \int_0^t r dt = \int_0^t \frac{1}{P(1 - \frac{P}{K})} \frac{dP}{dt} dt$$

$$\implies r(t - 0) = \int_{P(0)}^{P(t)} \frac{dP}{P(1 - \frac{P}{K})} \quad \text{change of variables}$$

$$\begin{aligned} \implies rt &= \int_{P_0}^P \frac{dP}{P(1 - \frac{P}{K})} \\ &= \int_{P_0}^P \left(\frac{1}{P} - \frac{1}{P - K} \right) dP \quad \text{partial fraction decomposition} \end{aligned}$$

$$= (\log |P| - \log |P - K|) \Big|_{P_0}^P = \log \left(\left| \frac{P}{P - K} \right| \right) \Big|_{P_0}^P$$

Logistic Model: exact solution (continued)

$$\begin{aligned}
 \Rightarrow \quad rt &= \log \left| \frac{P}{P-K} \right| - \log \left| \frac{P_0}{P_0-K} \right| \\
 &= \log \left| \frac{P}{P-K} \cdot \frac{P_0-K}{P_0} \right| \\
 &= \log \left(\frac{P}{P-K} \cdot \frac{P_0-K}{P_0} \right) \quad \text{\textit{P - K and P}_0 - K \text{ must have the same sign}} \\
 &= \log \left(\frac{1}{1-K/P} \cdot \frac{1-K/P_0}{1} \right) = \log \left(\frac{1-K/P_0}{1-K/P} \right)
 \end{aligned}$$

$$\Rightarrow e^{rt} = \frac{1-K/P_0}{1-K/P} \quad \Rightarrow \quad 1-K/P = (1-K/P_0)e^{-rt}$$

$$\Rightarrow \frac{K}{P} = 1 + (K/P_0 - 1)e^{-rt} \quad \Rightarrow \quad P = \frac{K}{1 + (K/P_0 - 1)e^{-rt}}$$

$$\Rightarrow P = \frac{P_0 K}{P_0 + (K - P_0)e^{-rt}} \quad \text{\textbf{SUCCESS!!}}$$

Logistic Model: exact solution

Suppose $0 < P_0 < K$, and suppose we want to know when the population reaches qK , where $P_0/K < q < 1$. Using

$$P(t) = \frac{P_0 K}{P_0 + (K - P_0)e^{-rt}},$$

set $P(t) = qK$ and solve for t :

$$\begin{aligned} qK &= \frac{P_0 K}{P_0 + (K - P_0)e^{-rt}} \\ e^{-rt} &= \frac{P_0(1 - q)}{q(K - P_0)} \\ t &= \frac{1}{r} \log\left(\frac{q(K - P_0)}{P_0(1 - q)}\right). \end{aligned}$$

Logistic Model: exact solution

What can we learn from the exact solution?

$$P(t) = P_0 \frac{K}{P_0 + (K - P_0)e^{-rt}}$$

- If $P_0 = 0$ then $P(t) \equiv 0$ (*equilibrium*)
- If $P_0 = K$ then $P(t) \equiv K$ (*equilibrium*)
- If $P_0 > 0$ and $P_0 \neq K$ then as $t \rightarrow \infty$, $P(t) \rightarrow K$
 - $\therefore K$ is a *stable* equilibrium (and 0 is *unstable*)
- If $0 < P_0 < K$ then as $t \rightarrow -\infty$, $P(t) \rightarrow 0$
- If $P_0 > K$ then there is some time $t_\star < 0$ when the solution $P(t)$ is undefined at $t_\star$ (!) so there is a solution only for times $t \in (t_\star, \infty)$. Should we be concerned about this model?
- Is $P_0 > K$ reasonable biologically? Could a population start above the carrying capacity?